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Designing a rigorous data collection and structuring methodology for human capital valuation

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Tristan de MARICOURT, Pavel-Dan DIACONU, Valentin GORSE, Nathan NOVIER

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Introduction

As AI systems become embedded in the fabric of major corporations, the conversation has shifted from whether to adopt them, to how to deploy them for maximum impact. Yet beneath this wave of adoption lies a more fundamental question, one that no algorithm has yet answered: what is the irreplaceable value of the human at work, and how can it be measured?

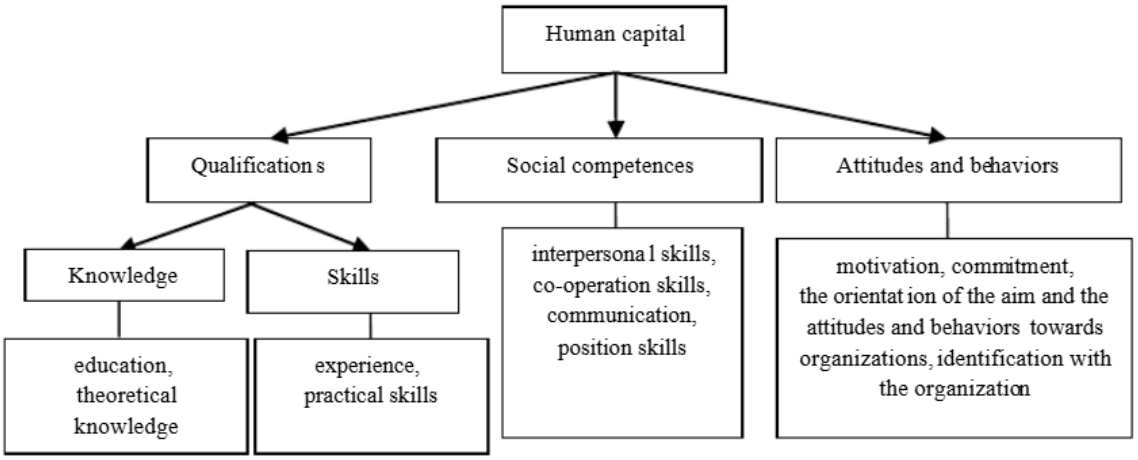
Here, we propose a structured framework for collecting and analysing data that enables organisations to assign quantifiable value to human contribution, not as a counterpoint to AI, but as a complement to it.

Our report is organised into three parts. The first establishes the theoretical framework underpinning our model for evaluating human value. The second examines the data required to operationalise this model, along with the methodologies for collecting it. The third, compiled in the annexe, surfaces a series of broader questions that, while beyond the scope of this summary, merit serious consideration.

1. Our model

1.1 Decomposing human capital

To operationalise the qualitative dimension outlined above, it is useful to draw on established human capital theory, which breaks individual value into three interconnected pillars. The first is **Qualifications**: encompassing both theoretical knowledge acquired through education and the practical skills built through experience. These form the technical foundation of an employee's contribution. The second pillar is **Social Competences**: interpersonal skills, communication, co-operation, and positional awareness. These qualities determine how effectively an individual's expertise is deployed within a team and across organisational boundaries. The third is **Attitudes and Behaviours**: motivation, commitment, goal orientation, and identification with the organisation. The third pillar acts as a multiplier on the first two: the same qualifications and social skills will produce vastly different outcomes depending on the degree of engagement behind them.



Schema of the theoretical decomposition of human capital, Guide on Measuring Human Capital, UN, 2016 (2)

Taken together, these three pillars offer a richer taxonomy for what we are attempting to measure. They also reveal why any single metric will inevitably fall short. What follows is therefore not a single model, but a layered set of approaches, each capturing a different dimension of human value, which can be combined and calibrated depending on the organisation's data maturity and analytical ambition.

1.2 From simple proxies to composite valuation

The most accessible starting point is the **EBITDA per Headcount** ratio: total EBITDA divided by the number of employees. Its strength lies in its universality. The inputs are standardised, externally reported, and immediately comparable across organisations, industries, and time periods. It should, however, be understood strictly as a benchmark rather than a diagnostic. By treating all employees as interchangeable units, it obscures the very differences in individual contribution, seniority, and function that this framework seeks to illuminate.

A more operationally grounded alternative is the **Productivity Ratio**, defined as Output / Hours Worked. Straightforward to track at both team and individual level, this measure is particularly valuable for detecting change over time, before and after a process improvement, a technology adoption, or an organisational restructuring. Its principal limitation is that it privileges volume over quality, and if used in isolation risks incentivising the wrong behaviours.

Moving further towards granularity, the **Net Employee Value** model attempts to isolate the true economic contribution of a single individual:

Net Employee Value = Value Produced + Fixed Cost Allocation – Salary – Transaction Costs

Here, **Value Produced** captures direct output attributable to the individual such as revenue generated, projects delivered, efficiency gains realised. **Fixed Cost Allocation** distributes shared overhead proportionally across the relevant headcount. **Transaction Costs** account for the frequently underestimated cost of turnover: recruitment, onboarding, and the productivity loss incurred during transition. **Salary** encompasses total compensation including benefits. Together, these terms begin to surface the full economic cost and return of an employee, well beyond what a pay slip reveals.

1.3 Towards a composite Human Capital Value score

While the models above offer useful approximations, none fully captures the multidimensional nature of human capital. The valuation of individual contribution within organisations has long resisted formalisation precisely because it is simultaneously technical, relational, market-dependent, and volitional. The framework proposed here addresses this by decomposing human capital value into four theoretically grounded and empirically tractable components, integrated into a single composite score.

The first, the **Qualification Score (Q)**, captures accumulated investment in education, certified training, and demonstrated performance against role-specific objectives:

$$Q = \alpha\ (\text{Education \& Certifications}) + \beta\ (\text{OKR of Achievement Rate}) + \gamma\ (\text{Time-to-Productivity at Hire})$$

The weighting coefficients α, β, and γ are calibrated by role family, drawing on **HRIS** (Human Resources Information System) records, **LMS** (Learning Management System) completion data, and **ATS** (Applicant Tracking System) sourcing metrics. This score establishes the technical baseline of an employee's contribution.

The second component, the **Behavioural and Collaborative Value Score (B)**, draws on social capital theory to quantify the relational and operational dimensions of individual contribution:

$$B = (\text{Network Centrality Score} \times \text{Collaboration Frequency}) + (\text{Project Velocity / Delivery Targets}) + (\text{Presence \& Mobility Index})$$

Network centrality, derived from organisational communication metadata, serves as a proxy for the degree to which an individual functions as a knowledge conduit and cross-functional integrator. This is precisely the **Transmission Value** dimension identified in the human capital decomposition above. Business outputs such as revenue per employee or **AuM** (Assets under Management) contribution provide an external validity check on the score.

The third component, the **Market-Adjusted Rarity Premium (R)**, introduces an externally referenced valuation consistent with the resource-based view of the firm:

$$R = \text{Salary Benchmark Ratio} \times \text{Role Scarcity Index} \times (1 + \text{Skill Inflation Factor})$$

This term captures what the labour market would demand to replace the individual. A high scarcity index combined with rising skill inflation signals that the employee's human capital is actively appreciating. Industry benchmarks from sources such as Mercer and Willis Towers provide the normalisation baseline.

The fourth component, the **Engagement and Attitude Multiplier (E)**, operationalises the engagement dimension of human capital:

$$E = \text{eNPS (employee Net Promoter Score)} \times \text{Sentiment Trend} \times (\text{Competency Map Coverage / Role Requirements})$$

Rather than treating motivation and organisational identification as inherently unquantifiable, this term draws on pulse survey sentiment, **NLP** (Natural Language Processing) processed exit interview signals, and the gap between self-declared and assessed competencies. Crucially, **E** enters the model not additively but as a multiplicative scalar. So it reflects the proposition that

attitudinal variables do not simply add to performance, but condition the extent to which qualifications and collaborative capacity are actually deployed. A highly qualified but disengaged employee will see their effective value discounted accordingly.

These four components are integrated into the **Composite Human Capital Value score**:

$$\text{HCV (Human Capital Value)} = (Q + B + R) \times E \times (1 - \text{Attrition Risk}) + \text{Succession Score}$$

The **Attrition Risk** term is generated by a predictive model trained on historical turnover data. It applies a discount proportional to the probability that the human capital stock will exit the organisation, acknowledging that value which cannot be retained is only partially realisable from the firm's perspective. The **Succession Score** captures the degree to which an individual's knowledge and capabilities are embedded in organisational routines and transferable to successor roles, accounting for the systemic rather than merely individual dimension of human capital.

This framework does not claim to resolve the fundamental epistemological tensions inherent in human capital measurement. Rather, it proposes a structured, data-grounded approximation that is both theoretically coherent and operationally implementable. It offers organisations a living, dynamic metric in place of the static and retrospective proxies that currently dominate people analytics practice.

2. Data collection

2.1 Data requirements by model component

The validity of the Human Capital Value framework is contingent on the quality, completeness, and ethical integrity of the data that feeds it. This section maps the data requirements of each model component onto concrete collection methodologies, cadences, and governance structures, translating the analytical framework into an operational blueprint.

The **Qualification Score (Q)** relies on educational and certification records from the HRIS and LMS, updated continuously; OKR achievement rates from the performance management system, refreshed quarterly; and time-to-productivity metrics from ATS and onboarding records, captured at hire and revisited upon internal mobility events.

The **Behavioural and Collaborative Value Score (B)** is the most technically demanding component, requiring communication and collaboration metadata from email, calendar, and enterprise messaging platforms to compute network centrality and collaboration frequency. Project velocity and delivery metrics are sourced from project management tools, and presence patterns from access and remote working records. Given the sensitivity of this data stream, only metadata should be processed, and never message content. Aggregation at team level is the recommended default before any individual-level analysis. Collection is continuous, scored monthly.

The **Market-Adjusted Rarity Premium (R)** draws exclusively on external data such as compensation benchmarks from providers such as Willis Towers Watson (4), that are updated annually. Role scarcity and skill inflation indices are constructed from labour market databases and job posting analysis tools. They're refreshed quarterly.

The **Engagement and Attitude Multiplier (E)** relies on primary employee data like eNPS and sentiment scores from pulse surveys administered monthly or bi-monthly, exit interview data processed through NLP pipelines, and competency assessments combining manager evaluations and self-declarations, conducted bi-annually. The voluntary and transparent nature of this collection must be unambiguous to employees at all times.

The **Attrition Risk and Succession Score** require longitudinal turnover data spanning a minimum of three to five years to train a reliable predictive model, complemented by competency maps and succession planning records maintained by HR business partners. The attrition model is retrained quarterly. Succession records reviewed bi-annually.

2.2 Governance principles

The data infrastructure underpinning the HCV framework operates under six non-negotiable governance principles. **Legality** requires that all collection comply with applicable data protection legislation, including for instance GDPR in European jurisdictions, with a documented legal basis for each data stream and mandatory legal review before activation. **Purpose limitation** restricts the use of HCV data exclusively to workforce analytics. It may not be repurposed for disciplinary decisions without a separate legal basis. **Data minimisation** mandates that only strictly necessary data be collected, with behavioural metadata retained for no longer than twelve months in identifiable form. **Transparency** requires that employees be informed in plain language of what is collected, how scores are derived, and what decisions they may influence, with full rights of access, correction, and human review. **Bias auditing** necessitates annual disaggregated analysis of composite scores across protected characteristics, conducted by an independent party, with any detected bias triggering mandatory model review. Finally, **human oversight** establishes that the HCV score is a decision-support tool only. No consequential talent decision may be made on its basis without documented human review and a dedicated governance committee to adjudicate edge cases.

2.3 Infrastructure and implementation

Implementation requires a centralised people analytics platform serving as the single system of record, with automated data quality pipelines, role-based access controls, and a complete audit log of data access and use. Data quality remediation, addressing gaps in HRIS records, OKR documentation, and LMS data, must precede any scoring rollout.

Deployment is best approached in three phases. First, infrastructure readiness and governance establishment. Second, a pilot of the simpler model components in a limited organisational unit. And third, a phased introduction of the full HCV score in an advisory capacity before any integration into talent management processes. Continuous employee communication throughout is a condition of the framework's legitimacy.

So the data architecture is organised across four layers, moving from structured internal systems to qualitative external signals, and is unified by a governance framework designed to ensure compliance, fairness, and auditability at every stage.

3. Annexes - Open questions and boundary conditions

The framework proposed in this report represents a structured first approximation of human capital valuation. Its responsible implementation, however, raises a series of questions that warrant serious consideration before any deployment. They are presented here not as obstacles, but as the necessary next frontier of this research agenda.

A. External conditions and seasonality

Any value metric measured over time is susceptible to confounding by macroeconomic cycles and seasonal patterns exogenous to individual performance. The framework must incorporate normalisation mechanisms such as rolling averages, market-adjusted benchmarking, and sector-specific deflators to ensure the HCV score reflects genuine contribution rather than favourable external conditions.

B. Bias and algorithmic fairness

A composite scoring model is only as neutral as the data it is trained on. And historical organisational data is rarely neutral. Structural biases relating to gender, ethnicity, age, or socioeconomic origin may be embedded in performance evaluations and network metrics in ways that are difficult to detect and easy to amplify. Rigorous bias auditing and independent ethics oversight are prerequisites for legitimate use.

C. Heterogeneity across functions

The coefficients and benchmarks appropriate for a research scientist are categorically different from those relevant to a sales executive or an engineer. Value creation in knowledge-intensive functions is often long-cycle and difficult to attribute. The model must be parameterised by function, seniority, and role family. A single universal score applied indiscriminately would reproduce precisely the reductionism this framework seeks to overcome.

D. Health-related absences

Where absenteeism results from work-induced conditions (burnout, chronic stress, or occupational injury) penalising the employee's HCV score would constitute a double injustice. Elevated rates of work-related absence should trigger an upward revision of organisational risk scores, not a downward revision of individual value.

E. Protection of vulnerable employees

The framework must incorporate explicit ethical guardrails for employees whose output is affected by circumstances beyond their professional conduct like serious illness, disability, or caregiving responsibilities. Protected variables must be excluded from scoring, and mandatory human review thresholds must precede any consequential talent decision informed by HCV data.

F. Wellbeing as a value driver

Investments in employee wellbeing are not merely ethical expenditures. They encompass nutrition, physical activity, mental health support, and flexible working. Evidence suggests they generate measurable returns in reduced absenteeism, lower attrition, and improved cognitive performance. A mature iteration of the HCV framework should incorporate a **Wellbeing Investment Return** coefficient, reframing employee health as one of the highest-return levers available to the organisation.

These questions define the conditions under which the framework can be applied responsibly, and constitute the natural agenda for the next phase of this research.

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